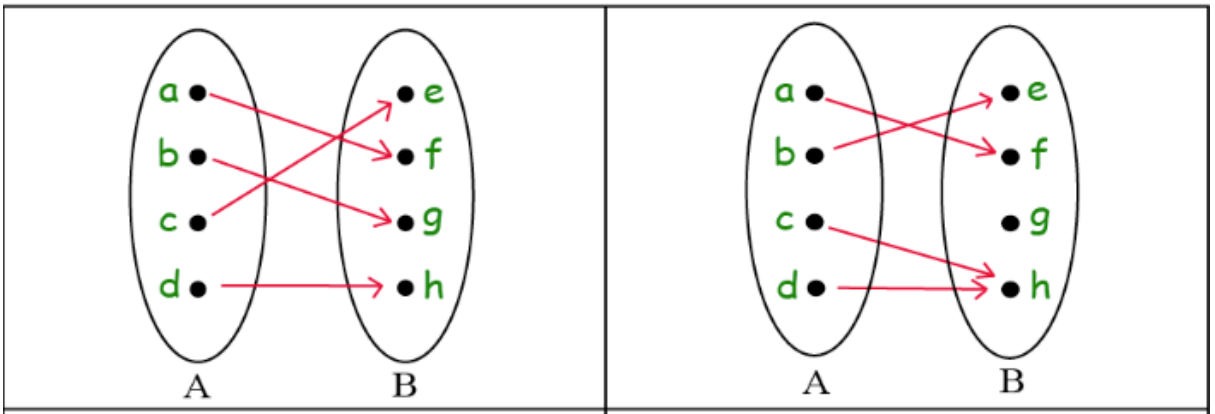


Surjective Examples

A function is said to be surjective if nothing in the codomain is left out, i.e. when the range = codomain.

Example 1



Yes, Surjective (onto)	Not Surjective (not onto)

Example 2

$$f : \{a, b, c, d\} \to \{1, 2, 3, 4\}$$
$$f = \{(a, 1), (b, 3), (c, 4), (d, 2)\}$$
$$=> \text{Yes, Surjective}$$

$$f : \{a, b, c, d\} \to \{1, 2, 3, 4\}$$
$$f = \{(a, 1), (b, 3), (c, 4), (d, 3)\}$$
$$=> \text{Not Surjective}$$

Example 3

$$f : \mathbb{R} \to \mathbb{R}, f(x) = e^x$$
$$\text{Not Surjective}$$
$$=> \text{Range: } y \in [0, \infty)$$
$$=> \text{Codomain: } y \in \mathbb{R}$$
$$(Range \neq Codomain)$$

Example 4

$$\text{Assume for any } y \in \mathbb{Z} \text{ there exists an } x \in \mathbb{Z} \text{ such that: } x = y + 8$$
$$\text{Then by the choice } x, \text{and definition of } f(x) \text{ this follows that:}$$
$$f(x) = (y + 8) - 8 = y$$
$$=> \text{y is an output of f and shows f is onto / surjective}$$